

Chapter 1 Examples and definitions of dynamical phenomena

A dynamical system can be any mechanism that evolves deterministically in time. Simple examples can be found in mechanics, one may think of the pendulum or the solar system. Similar systems occur in chemistry and meteorology. We should note that in biological and economic systems it is less clear when we are dealing with determinism.¹ We are interested in evolutions, that is, functions that describe the state of a system as a function of time and that satisfy the equations of motion of the system. Mathematical definitions of these concepts follow later.

The simplest type of evolution is stationary, where the state is constant in time. Next we also know periodic evolutions. Here, after a fixed period, the system always returns to the same state and we get a precise repetition of what happened in the previous period. Stationary and periodic evolutions are very regular and predictable. Here prediction simply is based on matching with past observations. Apart from these types of evolutions, in quite simple systems one meets evolutions that are not so regular and predictable. In cases where the unpredictability can be established in a definite way, we speak of chaotic behaviour.

In the first part of this chapter, using systems such as the pendulum with or without damping or external forcing, we give an impression of the types of evolution that may occur. We show that usually a given system has various types of evolution. Which types are typical or prevalent strongly depends on the kind of system at hand; for example, in mechanical systems it is important whether we consider dissipation of energy, for instance by friction or damping. During this exposition the mathematical framework in which to describe the various phenomena becomes clearer. This determines the language and the way of thinking of the discipline of Dynamical Systems.

In the second part of this chapter we are more explicit, giving a formal definition of dynamical system. Then concepts such as state, time, and the like are also discussed. After this, returning to the previous examples, we illustrate these concepts. In the final part of the chapter we treat a number of miscellaneous examples that regularly return in later parts of the book.

Here we particularly pay attention to examples that, historically speaking, have given direction to the development of the discipline. These examples therefore act as running gags throughout the book.

1.1 The pendulum as a dynamical system

As a preparation to the formal definitions we go into later in this chapter, we first treat the pendulum in a number of variations: with or without damping and external forcing. We present graphs of numerically computed evolutions. From these graphs it is clear that qualitative differences exist between the various evolutions and that the type of evolution that is typical or prevalent strongly depends on the context at hand. This already holds for the restricted class of mechanical systems considered here.

1.1.1 The free pendulum

The planar mathematical pendulum consists of a rod, suspended at a fixed point in a vertical plane in which the pendulum can move. All mass is thought of as being concentrated in a point mass at the end of the rod (see Figure 1.1), and the rod itself is massless. Also the rod is assumed stiff. The pendulum has mass m and length l . We moreover assume the suspension to be such that the pendulum not only can oscillate, but also can go ‘over the top’. In the case without external forcing, we speak of the free pendulum.

1.1.1.1 The free undamped pendulum

In the case without damping and forcing the pendulum is only subject to gravity, with acceleration g : The gravitational force is pointing vertically downward with strength mg and has a component $-mg \sin \varphi$ along the circle described by the point mass; see Figure 1.1. Here φ is the angle between the rod and the downward vertical, often called ‘deflection’, expressed in radians. The distance of the point mass from the ‘rest position’ ($\varphi = 0$), measured along the circle of all its possible positions, then is $l\varphi$. The relationship between force and motion is determined by Newton’s famous law $F = D ma$; where F denotes the force, m the mass, and a the acceleration.

By the stiffness of the rod, no motion takes place in the radial direction and we therefore just apply Newton’s law in the φ -direction. The component

of the force in the φ -direction is given by $-mg \sin \varphi$, where the minus sign accounts for the fact that the force is driving the pendulum back to φ . For the acceleration a we have

$$a = \frac{d^2(l\varphi)}{dt^2} = l \frac{d^2\varphi}{dt^2}$$

where $(d^2\varphi/dt^2)(t) = \varphi''(t)$ is the second derivative of the function $t \rightarrow \varphi(t)$. Substituted into Newton's law this gives

$$ml\varphi'' = -mg \sin \phi$$

or, equivalently,

$$\varphi'' = -\frac{g}{l} \sin \varphi$$

where obviously $m, l > 0$. So we derived the equation of motion of the pendulum, which in this case is an ordinary differential equation. This means that the evolutions are given by the functions $t \mapsto \varphi(t)$ that satisfy the equation of motion (1.1). In the sequel we abbreviate $w = \sqrt{g/l}$.

Observe that in the equation of motion (1.1) the mass m no longer occurs. This means that the mass has no influence on the possible evolutions of this system. According to tradition, as an experimental fact this was already known to Galileo,³ a predecessor of Newton concerning the development of classical mechanics.

Remark (Digression on Ordinary Differential Equations I). In the above example, the equation of motion is an ordinary differential equation. Concerning the solutions of such equations we make the following digression.

1. From the theory of ordinary differential equations (e.g., see [16, 69, 118, 144]) it is known that such an ordinary differential equation, given initial conditions or an initial state, has a uniquely determined solution.⁴ Because the differential equation has order two, the initial state at $t = 0$ is determined by the two data $\varphi(0)$ and $\varphi'(0)$, thus both the position and the velocity at the time $t = 0$. The theory says that, given such an initial state, there exists exactly one solution $t \mapsto \varphi(t)$, mathematically speaking a function, also called 'motion'. This means that position and velocity at the instant $t = 0$ determine the motion for all

future time.⁵ In the discussion later in this chapter, on the state space of a dynamical system, we show that in the present pendulum case the states are given by pairs (φ, φ') . This implies that the evolution, strictly speaking, is a map $t \mapsto (\varphi(t), \varphi'(t))$, where $t \mapsto \varphi(t)$ satisfies the equation of motion (1.1). The plane with coordinates (φ, φ') often is called the phase plane. The fact that the initial state of the pendulum is determined for all future time, means that the pendulum system is deterministic.⁶

2. Between existence and explicit construction of solutions of a differential equation there is a wide gap. Indeed, only for quite simple systems,⁷ such as the harmonic oscillator with equation of motion

$$\varphi'' = -w^2\varphi,$$

can we explicitly compute the solutions. In this example all solutions are of the form $\varphi(t) = A \cos(wt+B)$, where A and B can be expressed in terms of the initial state $(\varphi(0), \varphi'(0))$. So the solution is periodic with period $2\pi/w$ and with amplitude A; whereas the number B; which is only determined up to an integer multiple of 2π , is the phase at the time $t = 0$: Because near $\varphi = 0$ we have the linear approximation

$$\sin \varphi \approx \varphi$$

in the sense that $\sin \varphi = \varphi + o(|\varphi|)$, we may consider the harmonic oscillator as a linear approximation of the pendulum. Indeed, the pendulum itself also has oscillating motions of the form $\varphi(t) = F(t)$, where $F(t) = F(t+P)$ for a real number P; the period of the motion. The period P varies for different solutions, increasing with the amplitude. It should be noted that the function F occurring here, is not explicitly expressible in elementary functions. For more information see [153]. 3. General methods exist for obtaining numerical approximations of solutions of ordinary differential equations given initial conditions. Such solutions can only be computed for a restricted time interval. Moreover, due to accumulation of errors, such numerical approximations generally will not meet with reasonable criteria of reliability when the time interval is growing (too) long. 5

References

- [1] Henk Broer, Floris Takens, **Dynamical Systems and Chaos**, Springer, 2009.